

Assigning And Associating Elastic IP

Introduction to Elastic IPs

Elastic IPs (EIPs) are static public IPv4 addresses designed for dynamic cloud computing environments.

They allow you to mask the failure of an instance by remapping the address to another instance in your account.

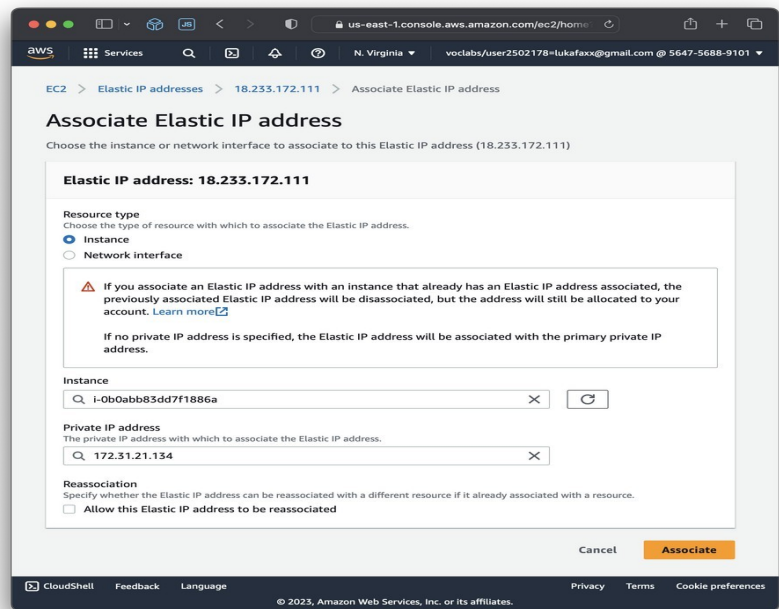
Understanding how to assign and associate Elastic IPs is crucial for maintaining high availability and reliable access to your resources.

Prerequisites for Assigning Elastic IPs

You need an active AWS account with proper permissions to allocate and associate Elastic IPs.

Make sure you have an existing EC2 instance or network interface to associate the Elastic IP with.

Familiarity with the AWS Management Console or AWS CLI is essential for efficient management.

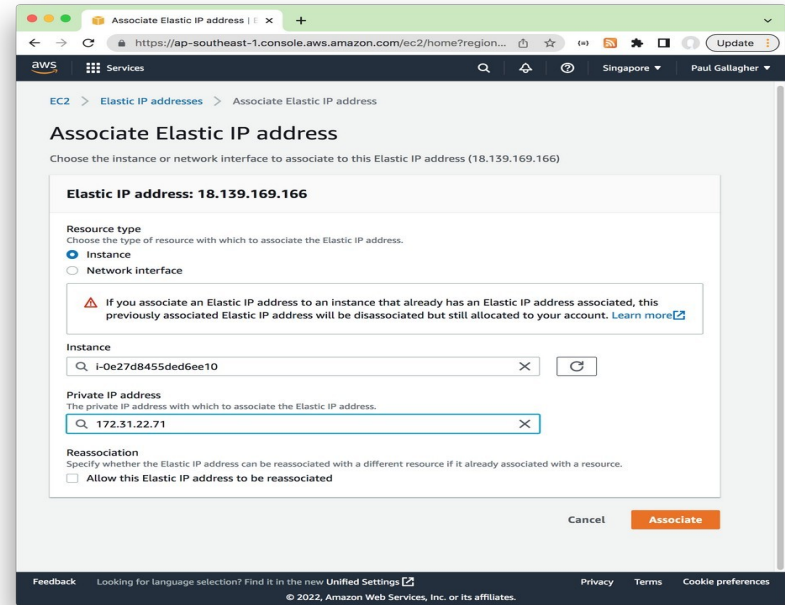


Allocating an Elastic IP via Console

Log in to the AWS Management Console and navigate to the EC2 Dashboard.

Select "Elastic IPs" from the network & security section and click "Allocate Elastic IP address."

Confirm the allocation, choose the appropriate scope (VPC or EC2-Classic), and click "Allocate."



The screenshot shows the AWS Management Console interface for associating an Elastic IP address. The breadcrumb navigation indicates the path: EC2 > Elastic IP addresses > Associate Elastic IP address. The main heading is "Associate Elastic IP address", followed by the instruction: "Choose the instance or network interface to associate to this Elastic IP address (18.139.169.166)".

The "Elastic IP address: 18.139.169.166" is displayed. Under "Resource type", the "Instance" radio button is selected. A warning message states: "If you associate an Elastic IP address to an instance that already has an Elastic IP address associated, this previously associated Elastic IP address will be disassociated but still allocated to your account. [Learn more](#)".

The "Instance" field contains the ID "i-Oe27d845ded6ee10". The "Private IP address" field contains "172.31.22.71".

Under the "Reassociation" section, there is a checkbox labeled "Allow this Elastic IP address to be reassociated", which is currently unchecked.

At the bottom right, there are "Cancel" and "Associate" buttons. The footer includes a "Feedback" link, a language selection prompt, a link to "Unified Settings", and copyright information for Amazon Web Services, Inc. or its affiliates.

Associating Elastic IP to an Instance

Go to the "Elastic IPs" section, select the desired IP, and click "Actions" > "Associate address."

Choose the instance or network interface you want to associate the IP with from the dropdown menus.

Click "Associate" to complete the process, making the Elastic IP reachable from the internet.

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (18.209.53.97)

Elastic IP address: 18.209.53.97

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance
☐ Network interface

Warning If you associate an Elastic IP address with an Instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance

i-0923e2bde326445e5 (MyWebServer) - running

The private IP address with which to associate the Elastic IP address

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

☐ Allow this Elastic IP address to be reassociated

Associating Elastic IP using AWS CLI

Use the command `aws ec2 associate-address`` with parameters for the IP and instance ID.

Example: `aws ec2 associate-address --allocation-id eipalloc-12345678 --instance-id i-1234567890abcdef0``.

This method provides automation and scripting capabilities for large-scale deployments.

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (18.209.53.97)

Elastic IP address: 18.209.53.97

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☒ Instance
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Warning If you associate an Elastic IP address with an Instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance
Choose an Instance
i-0923e2bde326445e5 (MyWebServer) - running
The private IP address with which to associate the Elastic IP address

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
☐ Allow this Elastic IP address to be reassociated

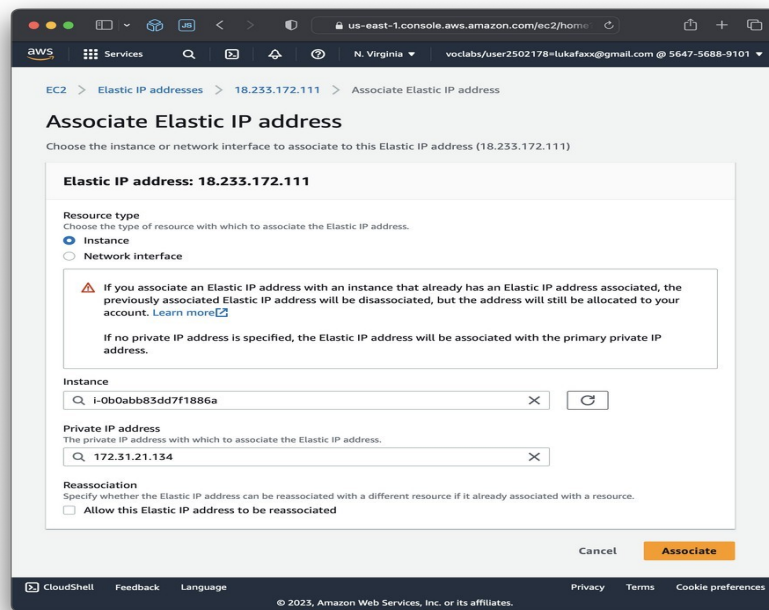
Cancel Associate

Reassociating and Releasing Elastic IPs

To re-associate an Elastic IP, first disassociate it from the current resource, then associate it with a new resource.

To release an Elastic IP, select it in the console and click "Release addresses," ensuring it is no longer in use.

Proper management of Elastic IPs helps avoid unnecessary charges and resource conflicts.



Best Practices for Using Elastic IPs

Use Elastic IPs sparingly, as AWS charges for idle IP addresses not associated with a running instance.

Always disassociate Elastic IPs before terminating instances to prevent additional charges.

Automate IP management with scripts or infrastructure as code tools for consistency and efficiency.

Summary and Key Takeaways

Assigning and associating Elastic IPs is essential for maintaining consistent public access to cloud resources.

Proper management includes allocation, association, disassociation, and release practices.

Leveraging AWS tools and best practices ensures efficient, cost-effective, and reliable IP address management.

